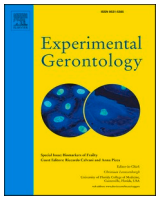
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Investigating the effect of ambient temperature, sound type, and noise level on the heart rate of the elderly

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ABSTRACT

Background: Given the increasing elderly population in the future, attention to the health of older adults is important. Environmental factors, including ambient temperature, sound type, and noise level, greatly affect the health of older adults. This study looks at how these elements influence the heart rate of elderly individuals.
Method: This study involves laboratory research with 216 elderly male participants from Ilam province in Iran over four months, from May to August 2024. Participants were exposed to different ambient temperature settings (20 °C, 25 °C, and 30 °C) and four noise levels (55, 65, 75, and 85 dB(A)) using sound types like music, traffic, and crowded noise. They were randomly divided into three equal groups of 72 by ambient temperature and then into subgroups of 24 based on the three sound type; finally, these subgroups were tested under four different noise levels, with each group consisting of 6 participants. The data from participants were analyzed using a three-way ANOVA model.
Results: The findings showed that ambient temperature, sound type, and noise level greatly affected older adults' heart rate, with strong statistical significance. There was a notable interaction between ambient temperature and noise level. However, other interactions did not show significant relationships with heart rate.
Conclusion: The study highlights important impacts on the health of elderly in noisy urban areas. It shows that long exposure to high noise and heat increases heart rate. This research emphasizes noise control and heat management to protect older adults and informs public health policies and urban planning.

1. Introduction

Population aging is a significant demographic challenge and a key driver of social change in the 21st century, with its effects expected to increase in the coming years (Rowland, 2012). The growing elderly population, especially "older adults," is recognized globally as a critical issue that is placing increasing pressure on health services, social care, housing, and the economy (Christensen et al., 2009). Population aging is characterized by an increase in the number and proportion of individuals aged 60 years and older, along with a decline in the number and percentage of those aged 15 years and younger (Ismail et al., 2021). Population aging is occurring in both developed and developing countries, although it is most rapid in newly industrialized and developing regions (Noroozian., 2012). Many scholars argue that population aging has emerged as a significant demographic phenomenon that warrants careful consideration (Cristea et al., 2020).

The rising aging population in Iran is increasing more quickly than the global average. Iran has the second fastest-growing older population in the world, with a significant rise in the percentage of people aged over 60, projected to reach 14. 8 % by 2030 and 29. 4 % by 2050 (Noroozian., 2012). The aging population in Iran is growing at a rate that exceeds the global average. According to data from the Statistical Center of Iran, looking at the elderly population of Iran (65 years and older), it can be observed that the elderly population of Iran has increased from about 1.6 % of the total population in 2016 to 7.8 % in 2024. Also, based on the center's forecasts, over 10.4 million people will be added to the population of Iran over the 20 years from 2016 to 2036, among which the elderly population (65 years and older) will increase by about 5.5 million people or equivalently 11.5 % (Statistical Centre of Iran, 2024).

To improve the health and well-being of older adults in Iran, collaboration between the government, private sector, civil society, and the elderly is essential. Establishing geriatric assessment units and comprehensive care programs is recommended, along with increasing the number of geriatric specialists. It is important to focus on the

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distinctive needs of older adults and make necessary policy adjustments, allocate resources, and enhance healthcare services specific to this demographic (Piroozi et al., 2024). Policymakers and healthcare managers are encouraged to invest in legislative measures, prevention strategies, and resource allocation that effectively respond to the needs of the elderly. By fostering partnerships within the community and enhancing the current health infrastructure, the adaptability and responsiveness of healthcare systems can be improved (Samouei et al., 2024). The global increase in life expectancy emphasizes the importance of understanding the factors that influence the well-being of older adults (Crimmins and Zhang, 2019). Age is widely recognized as a critical factor in thermal comfort perception (Wu et al., 2023; Jiao et al., 2023). Due to disruptions in body ambient temperature regulation (Baquero and Forcada, 2022; Inoue et al., 1991), older adults do not respond to hot and cold environments as effectively as younger individuals (Al-Absi et al., 2022; Zhu et al., 2018). As ambient temperatures rise, older adults often experience an increase in heart rate (Zhu et al., 2018). A variety of policies have been implemented to enhance the quality of life for older adults in urban areas, with the auditory environment being a significant factor affecting overall comfort (Baquero Larriva and Higueras García, 2023).

Noise, defined as unwanted sound, poses potential risks to health and communication (Ismaila and Odusote., 2014). Noise-induced stress negatively impacts human performance (Simpson et al., 1974), and is associated with neurological and psychological complaints (Baquero Larriva and Higueras García, 2023). It is recognized as one of the four major pollutants that severely threaten human physical and mental health. Factors such as construction, traffic, and crowd noise create fluctuating sound pressure levels due to their random occurrence (Guan et al., 2020). Consequently, exposure to traffic noise can influence heart rate variability (Huang et al., 2013), and crowd noise is linked to increased heart rates (El Aarbaoui and Chaix, 2020).

Individuals over 65 years old are at a greater risk for heart disease due to noise exposure compared to younger individuals (Vienneau et al., 2015). Therefore, noise serves as an environmental factor that elevates heart rate (Schnell et al., 2013). Heart rate is one of the most important indicators of cardiovascular health and holds special significance for the elderly. There are substantial knowledge gaps between the biological aging hallmarks and aging-related diseases in humans (Zhao et al., 2024). As age increases, physiological changes occur in the cardiovascular system that can have significant effects on the well-being of elderly individuals (Al-Tabtabai et al., 2025).

Monitoring heart rate in older adults can aid in the early diagnosis of issues such as arrhythmias, heart failure, or coronary artery diseases. Additionally, controlling heart rate and maintaining it within a normal range can help prevent serious complications such as heart attacks or strokes (Ndlovu and Chungag, 2024). This is especially important for the elderly, who are more at risk for these dangers. Research on transportation noise as a risk factor for cardiovascular death was summarized in a 2021 meta-analysis (Cai et al., 2021). Learning stress management techniques and reducing heart rate during stressful times can contribute to the mental health improvement of the elderly. Understanding the impact of heat stress on the cardiovascular system in older adults underscores the importance of developing adaptation strategies, interventions and policies to protect this vulnerable population during extreme heat events (Liu et al., 2025). Overall, paying attention to heart rate and cardiovascular health in older adults not only helps to preserve their health but can also enhance their quality of life (Jakovljevic, 2018). A collaborative approach involving various stakeholders, including researchers, is necessary to improve geriatric care. Sharing successful international care models can provide insights for better practices in Iran, specifically in Ilam province.

The present study aims to examine the influence of environmental factors, including ambient temperature, sound type, and noise level on the heart rate of elderly individuals. Given the aging population and the increased susceptibility of older adults to environmental factors, understanding how ambient temperature and noise affect their heart rate can lead to improved care measures. For the health and well-being of older adults, collaboration among the government, private sector, and civil society is crucial. Recommendations include establishing geriatric assessment units and enhancing healthcare services. Policymakers should invest in legislation, prevention strategies, and community partnerships to improve healthcare systems for the elderly.

2. Methods

2.1. Study participants

This study was approved by the Ethics Committee of the Islamic Azad University, Ilam Branch, with the identification code IR.IAU.ILAM.RES.1403.012. A total of 216 healthy elderly men, aged between 65 and 75 years, with a normal body mass index, were selected for the study and agreed to participate. In this study, relative humidity was set at 50 %, and airspeed was <0.1 m per second. Three ambient temperature levels (20, 25 and 30 °C) corresponding to cool, neutral, and warm sensations were tested, ensuring that the conditions were stable for 10 h before the experiment. Additionally, four noise levels (55, 65, 75, and 85 dB denoted as dB(A)), reflecting typical exposure levels of noise throughout the day, and three types of sound (music, traffic, and crowded) were tested. The metabolic rate was approximately 1.0 MET, and the clothing insulation value was 0.6 Clo. For each participant, standard clothing with the same material and design was provided to ensure that the clothing coverage rate was consistent across individuals.

To assess the eligibility of individuals with a history of cardiovascular diseases or sensory-motor and auditory disorders, we initially verified the qualifications of the older adults participating in this study through interviews and self-reporting. Participants were assured that their involvement in the research was entirely voluntary and were instructed not to include any identifying information on the questionnaire. The eligibility criteria for participants included: - Full consent to undergo the relevant tests - An age range of 65 to 75 years - Mental alertness - Cardiovascular health - Absence of auditory and sensory-motor disorders. Additionally, individuals were excluded from the study if they had issues with alertness or any auditory or sensory-motor disorders that could interfere with their experience in a thermal environment. After confirming participants' eligibility, they were asked to stay at a designated location, free from noise, the day before the tests.

The experiment method in this study is as follows: the participants, who include 216 elderly men, were randomly divided into three equal groups of 72 people based on the environmental ambient temperature variable at three ambient temperature levels: 20 °C, 25 °C, and 30 °C. Then, each of the three groups mentioned above was also randomly divided into three equal groups of 24 people to measure the environmental sound type variable (music, traffic, and crowd noise); finally, each of these 24-person groups was tested under four noise levels (55, 65, 75, and 85 dB(A)). It is important to note that in this study, the heart rate of the participants was not measured before and after the experiment because to examine the heart rate of the elderly participants, we first placed them at a neutral temperature of 25 °C for 20 min to control individual tensions and mental and psychological factors that might affect the results of the experiment, and for the elderly to reach thermal adaptation. In addition, their heart rate was not recorded in the pre-test phase and before entering the laboratory chamber to reduce the sensitivity of the elderly to heart rate measurement under laboratory conditions and minimize psychological effects on the experiment results. Further details on how the tests were conducted are provided below.

2.2. Measurements

2.2.1. Laboratory equipment

The experiments were conducted in a climate chamber that consists of three sections: changing room, a room with an ambient temperature

of 25 °C, which is considered comfortable, and a laboratory room (Fig. 1). Noise stimuli were delivered through a speaker placed at the center of the table. Environmental parameters, such as ambient temperature and relative humidity, were measured using a data logger (delta log 10). This device records various climatic variables, including ambient temperature, humidity, airflow velocity, radiant ambient temperature, and relative humidity. Noise levels were assessed with a noise meter that measures frequencies ranging from 10 Hz to 20,000 Hz, with a maximum capacity of 140 dB(A). In the current experiment, the ambient temperature and humidity of the climate chamber were controlled by the data logger, which maintained a margin of error of ±0.3 degrees for three hours before the tests.

To assess the potential impact of noise on physiological parameters, participants measured their heart rate using a small desktop device (Omron, HEM-7320), which can measure rates from 40 to 180 beats per minute with an accuracy of ±5 %. Participants conducted the heart rate test themselves and then recorded the results on the questionnaire form. After a half-hour of thermal adaptation, participants were placed in one noise level environment for 15 min. It is crucial to present the samples to participants in a counterbalanced manner to avoid order effects. The sound was played in a randomized sequence during the experiment. Following this, the operating staff adjusted the noise level. The entire experimental process lasted 120 min, maintaining a constant thermal environment.

2.2.2. How to perform the test

This study investigated the combined effects of four noise levels (55, 65, 75, and 85 dB(A)), ambient temperatures (20 °C, 25 °C, and 30 °C), and sound types (music, traffic, and crowded noise) on the heart rates of elderly men. A total of 216 healthy elderly men, aged 65 to 75 years, with a normal body mass index, were selected for the experiment. Participants were instructed to abstain from caffeine and smoking for 24 h before the study. Additionally, their hearing was tested using an audiometer to ensure none had hearing impairments. The 216 participants were initially divided into three main groups of 72 individuals each, with each group exposed to a specific type of sound: music, traffic, or crowded noise. Each of these main groups was further subdivided into three subgroups of 24 participants, each tested under different ambient temperature conditions (20 °C, 25 °C, and 30 °C). Within each ambient temperature condition, participants were exposed to four different noise levels (55, 65, 75, and 85 dB(A)).

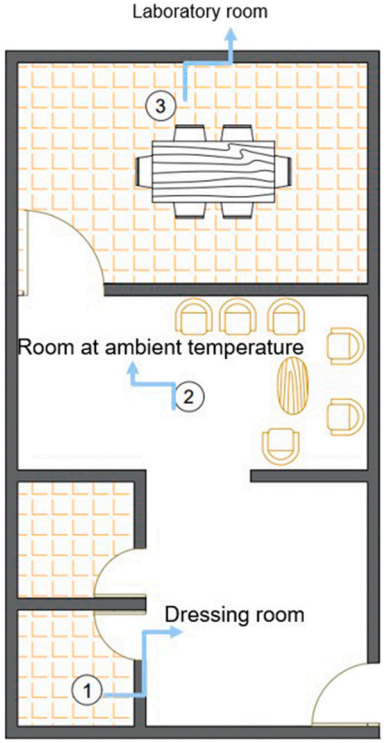


Fig. 1. Clinical trial environment design.

To facilitate the experiments, each group of 24 was divided into two groups of 12, and then into smaller groups of 6. Each testing session involved 6 participants. Participants entered the chamber room. After a half-hour of thermal acclimation, they were exposed to a specific noise-level environment for 15 min. During this time, they completed questionnaires. Upon arrival, participants spent 5 min in room 1 changing into the provided standardized clothing. Next, in room 2, they waited for 20 min in a pre-conditioning chamber maintained at 25 °C with 50 % relative humidity while wearing the experimental clothing to ensure uniform thermal conditions. During this time, the experiment and questionnaires were explained to them. Following that, in room 3, where the desired ambient temperature was set 10 h earlier, they are placed for 15 min under the specified ambient temperature and noise levels. Finally, at the last step, which took 80 min to execute, participants experienced four 20-min intervals, during which they were exposed to different noise levels for 15 min, and 5 min were spent filling the questionnaires and recording individuals' heart rates. Fig. 2 shows the schedule of each test.

3. Results

Statistical analyses were performed using SPSS 26. The analysis of variance (ANOVA) was used in this study. The level of significance was fixed at $P = 0.05$. The mean and standard deviation of the variables for each group are displayed in (Table 1). The mean heart rate of elderly individuals varied based on ambient temperature levels, recorded at 20 °C, 25 °C, and 30 °C, with values of 91.67, 69.65, and 80.63 beats per minute, respectively. The highest mean heart rate, averaging 91.50 beats per minute, was observed in the crowd noise group at an ambient temperature of 30 °C and a noise level of 85 dB(A). In contrast, the lowest mean heart rate, recorded at 63.50 beats per minute, was associated with the music sound type group at an ambient temperature of 20 °C and a noise level of 55 dB(A). Results also showed a significant interaction effect between temperature and noise level on heart rate; $p < 0.05$. However, the interaction effects between temperature and sound type $p > 0.05$, sound type and noise level, and the three-way interaction between temperature, sound type, and noise level were not significant; $p > 0.05$. Also, Figs. 3–5 illustrate the two-way and simultaneous interaction effects of temperature, noise level, and sound type on the heart rate of elderly individuals under different temperature conditions, showing detectable significant effects. Of course, readers can refer to (Appendix A) for more information and to see the results of the analysis of variance of the heart rate of the elderly participants in this study.

One of the key assumptions in performing an analysis of variance (ANOVA) is that the error variance of the dependent variable is uniform across the study groups. In this case, Levine's statistic is 2.161. At the 1 % significance level, we do not have sufficient evidence to reject the assumption of equal variances among the groups, as $p > 0.01$. Consequently, we can proceed with the analysis of variance to examine the differences between the study groups.

According to (Fig. 1), varying noise levels have different effects on heart rate at different ambient temperatures. Therefore, by controlling for the variable of sound type (specifically music), a two-way analysis of variance was performed only on ambient temperature and noise level. The results indicate significant effects for ambient temperature levels; $p < 0.01$ and sound type; $p < 0.01$ on heart rate. Additionally, there is a significant interaction effect between ambient temperature and noise level; $p < 0.05$. On the other hand, the results from Tukey's test indicate a significant difference in the mean heart rate among elderly individuals exposed to different noise levels. Specifically, there is a significant

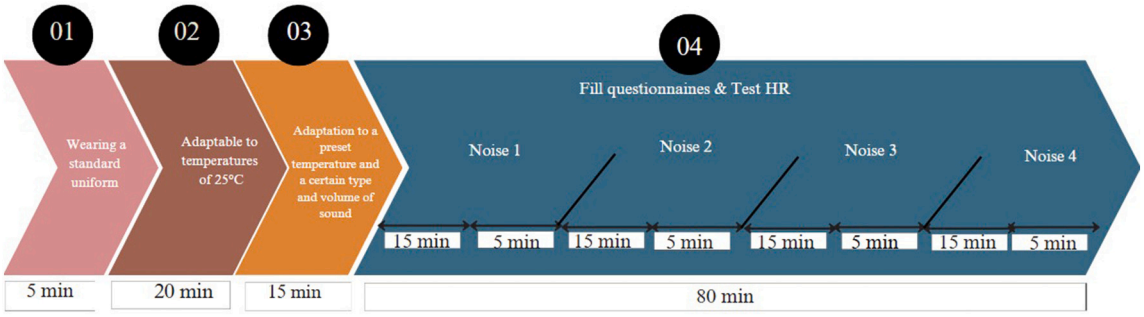


Fig. 2. Schedule of test.

Table 1
Means and standard deviations of heart rate of elderly people (mean ± SD).

Sound type factor levels	Noise levels	Ambient temperature factor levels			Statistical significance
		Ambient temperature 20 °C	Ambient temperature 25 °C	Ambient temperature 30 °C	Temperature factor levels Sound type factor levels Noise level factor levels (p-Value = 0.001)
The sound of music	55 dB(A)	63.1±5.1	66.1±8.4	74.3±1.7	The interaction effect of the temperature factor on the noise level factor (p-value = 0.001)
	65 dB(A)	64.1±0.1	67.1±6.7	74.3±6.6	
	75 dB(A)	66.1±5.1	69.2±5.1	81.3±6.6	The interaction effect of the temperature factor on the sound type factor (p-value = 0.084)
	85 dB(A)	68.2±3.3	73.2±1.63	85.3±0.1	
Traffic noise	55 dB(A)	66.1±8.4	67.2±0.0	75.3±6.7	The interaction effect of the sound type factor on the noise level factor (p-value = 0.814)
	65 dB(A)	67.1±1.6	67.1±8.3	75.3±6.6	
	75 dB(A)	69.2±0.1	69.2±8.4	83.4±8.3	Interactions between temperature factors, sound type factor, and noise level (p-value = 0.998)
	85 dB(A)	71.1±6.7	73.2±6.7	88.5±1.7	
The sound of the crowded	55 dB(A)	67.1±1.7	67.1±5.1	76.4±1.6	
	65 dB(A)	68.1±0.0	68.1±3.4	76.3±1.7	
	75 dB(A)	69.3±6.6	70.2±1.7	85.4±0.1	
	85 dB(A)	73.2±1.7	74.3±3.3	91.6±5.1	
Total		67.3±9.1	69.3±6.5	80.6±6.4	

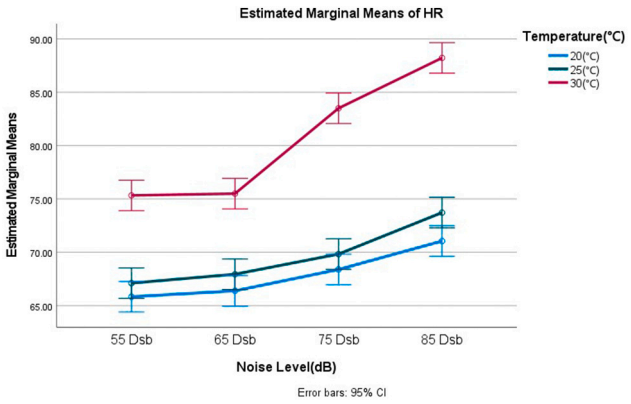


Fig. 3. Interaction effects between different levels of air ambient temperature and noise levels on the heart rate of elderly people.

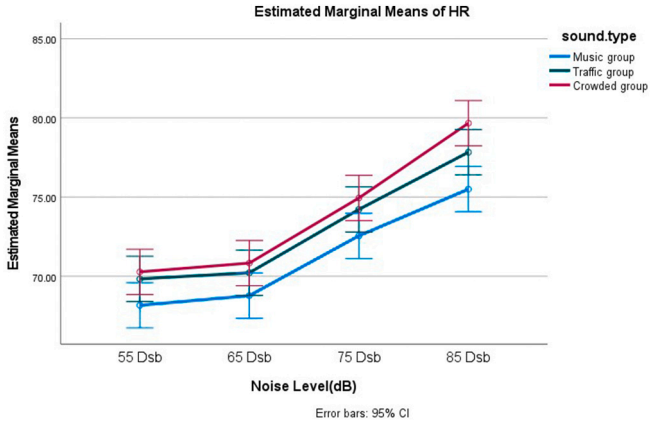


Fig. 4. Diagram of the interaction effects between different levels of sound type and sound noise level on the heart rate of elderly people.

difference between the noise levels of 75 dB(A) and 85 dB(A) compared to the group exposed to 55 dB(A); $p < 0.05$. However, no significant difference was found between the 65 dB(A) and 55 dB(A) groups; $p > 0.05$. This finding is illustrated in (Table 2), which categorizes the sound-level groups into subgroups that reflect their similarities. In this table, the mean noise level groups of 55 dB(A) and 65 dB(A) are categorized separately from the group exposed to 75 dB(A) and 85 dB(A), which do not differ significantly from each other.

3.1. Two-way variance with variable control of music sound type

By controlling the variable of noise levels, specifically the level of music, and conducting a two-way analysis of variance on the factors of ambient temperature and noise level, the results demonstrate significant effects of both ambient temperature and noise level on the heart rate of elderly individuals. This is indicated by the values: $F(2.60) = 169.215$, $p < 0.01$ for ambient temperature, and $F(3.60) = 31.821$, $p < 0.01$ for noise level. Additionally, the interaction effect of ambient temperature and noise level on the heart rate of elderly individuals was significant, with a value of $F(6.60) = 2.701$, $p < 0.05$. Furthermore, Tukey's test results revealed a significant difference in the mean heart rates of elderly individuals exposed to different noise levels. Notably, there was a significant difference between the 75 & 85 dB(A) noise level groups compared to the group exposed to 55 dB(A); $p < 0.05$. However, no

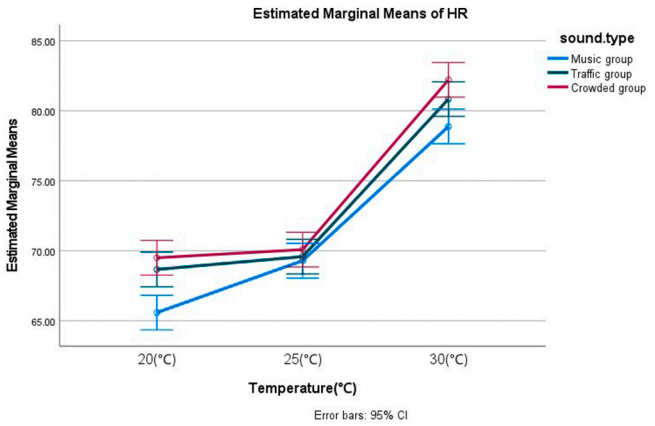


Fig. 5. Diagram of the interaction effects between different levels of sound type and air ambient temperature on the heart rate of elderly people.

Table 2
Homogeneous subsets with the control variable of music sound type.

Homogeneous subsets			
Set 1	Set 2	Set 3	
(68.2)[a] for noise level group 55 dB(A) (68.8)[a] for noise level group 65 dB(A)	(72.6)[a] for noise level group 75 dB(A)	(75.5)[a] for noise level group 85 dB(A)	
0.893	**0.877**	**0.757**	p-Value

The bold values given in the table meant $p > 0.05$ and there were no significant difference.

[a] The numbers relate to the comparison of means between two groups, which is only significant for set 1.

significant difference was found between the group exposed to 65 dB(A) and 55 dB(A); $p > 0.05$. This data is illustrated in (Table 2), which categorizes the groups into subgroups that are consistent with each other. In this table, the means of the 55 dB(A) and 65 dB(A) noise level groups differ from the means of the group exposed to 75 dB(A) and 85 dB (A). The latter two groups are placed in a separate subgroup, indicating that they do not differ significantly from each other.

The numbers in (Table 2), represent the average heart rates of elderly participants in this study across four noise levels (55, 65, 75, and 85 dB (A)). These averages have been categorized into three sets based on their same mean values denoted by: Set 1, Set 2, and Set 3. The average heart rates for the 55 dB(A) and 65 dB(A) groups demonstrated significant difference from the other groups. This categorization process has been applied consistently in the subsequent tables presented. As stated, the main goal of the authors is to investigate the effect of three independent variables: ambient temperature, sound type, and noise level, on the dependent variable (heart rate) of elderly. The results of the significant interaction effect of the variables are presented in (Table 1). Of course, the complete results of the analysis of variance are also provided.[1] In addition, the researchers are seeking to examine whether, by factoring out and removing the variable of sound type, the other two variables, temperature and noise level, still affect individuals' heart rate. For this reason, since there is no significant difference between the 55 dB(A) and 65 dB(A) groups; $p > 0.05$, so they are jointly placed as a subset in Set 1. On the other hand, since for all three sets (Set1, Set2, and Set3) the p-value is >0.05, there is no significant difference between these sets in terms of their effect on elderly heart rate.

[1] Additional details, including the ANOVA table for the heart rates of participants in this study and diagrams related to the interaction effects between variables, can be found in Appendix A.

3.2. Two-way variance with variable control of traffic sound type

By controlling for traffic noise levels and conducting a two-way analysis of variance solely on the factors of ambient temperature and noise level, the results show that both ambient temperature and noise level have a significant effect on the heart rate of elderly individuals. Additionally, the results from Tukey's test indicated a significant difference in average noise levels among elderly individuals in groups exposed to noise levels of 75 dB(A) and 85 dB(A) when compared to the group exposed to 55 dB(A); $p < 0.05$. However, no significant difference was found between the average noise levels in the group exposed to 65 dB(A) and the group exposed to 55 dB(A); $p > 0.05$. This information is detailed in (Table 3), which categorizes the groups into homogeneous subsets. In this table, the averages for the group exposed 55 dB(A) and 65 dB(A) are placed in a separate subset from the group exposed to 75 dB (A) and 85 dB(A), indicating that the latter two do not differ significantly from each other.

3.3. Two-way variance with variable control of crowd sound type

By controlling for the variable of noise level (crowd noise) and conducting a two-way analysis of variance solely on the factors of ambient temperature and noise level, the results indicate that both ambient temperature and noise level, as well as their interaction on the heart rate of the elderly, are significant. The respective values are ($F(2.60) = 98.944$, $p < 0.01$), ($F(3.60) = 27.326$, $p < 0.01$), and ($F(6.60) = 3.644$, $p < 0.01$). Additionally, Tukey's test results showed a significant difference in the average noise levels among elderly individuals between the groups with noise levels of 75 & 85 dB(A) and the group with 55 dB(A); $p < 0.05$. However, no significant difference was observed between the average noise level in the group with 65 dB(A) and the group with 55 dB(A); $p > 0.05$. This result is shown in (Table 4), where the groups are divided into homogeneous subsets. In this table, the average of the 55 dB(A) and 65 dB(A) groups is placed in a separate subset from the groups exposed to 75 dB(A) and 85 dB(A), as the latter two groups do not have a significant difference between themselves.

Fig. 3 shows the interaction between ambient temperature and noise level on heart rate in elderly people. Three ambient temperature conditions (20 °C, 25 °C, 30 °C) with noise levels (55 dB(A), 65 dB(A), 75 dB (A), 85 dB(A)) were investigated. In general, heart rates at 20 °C and 25 °C were almost the same, but heart rates at 25 °C were slightly higher than at 20 °C. At both 20 °C and 25 °C, heart rate values were similar at noise levels of 55 dB(A) and 65 dB(A), and the values at 75 dB(A) and 85 dB(A) were also similar.

Fig. 4 shows the interaction effects between different sound type and noise level on heart rate in older adults. Three types of sound (music, traffic, and crowded) were examined at different noise levels (55 dB(A), 65 dB(A), 75 dB(A) and 85 dB(A)). Overall, heart rates were similar in

Table 3
Homogeneous subsets with variable control of traffic sound type.

Homogeneous subsets			
Set 1	Set 2	Set 3	
(69.8) for noise level group 55 dB(A) (70.2) for noise level group 65 dB(A)	(74.2) for noise level group 75 dB(A)	(77.8) for noise level group 85 dB(A)	
0.980	**0.635**	**0.212**	p-Value

The bold values given in the table meant $p > 0.05$ and there were no significant difference.

Table 4
Homogeneous subsets with the control variable of the crowd sound type.

Homogeneous subsets			
Set 1	Set 2	Set 3	
(70.3) for noise level group 55 dB(A) (70.8) for noise level group 65 dB(A)	(74.9) for noise level group 75 dB(A)	(79.7) for noise level group 85 dB(A)	
0.965	**0.835**	**0.723**	p-Value

The bold values given in the table meant $p > 0.05$ and there were no significant difference.

traffic and crowded at 55 dB(A) and 65 dB(A). However, at 75 dB(A) and 85 dB(A), heart rates increased in proportion to traffic and crowded, but crowdedness consistently increased heart rates relative to traffic. Conversely, music tended to maintain lower heart rates than traffic and crowded at all noise levels tested.

Fig. 5 illustrates the interaction effects of different sound types and air ambient temperatures on heart rates in older adults. The study examined three types of sound (music, traffic, and crowded), under three ambient temperature conditions: 20 °C, 25 °C, and 30 °C. Overall, heart rates were similar for traffic and crowd sounds at both 20 °C and 25 °C. However, music consistently produced lower heart rates. At 30 °C, both traffic and crowd sounds caused an increase in heart rates compared to music.

In this study, we focused on investigating the impact of two variables—noise level and ambient temperature—on heart rate while keeping the "type of sound" constant. By limiting our analysis to a single type of sound each time, we aimed to determine whether noise level and ambient temperature significantly affect individuals' heart rates. The necessary analyses have been completed, providing detailed explanations about the simultaneous interaction effects of three sound types—music, traffic, and crowd noise—along with different levels of ambient temperature and noise levels on the heart rates of elderly individuals. This includes relevant charts to illustrate the findings.[2]

4. Discussion

The present study demonstrates that both ambient temperature and noise level significantly affect heart rate in elderly individuals. Elderly care is a crucial issue influenced by environmental factors such as ambient temperature, sound type, and noise level, garnering the attention of many researchers (Verderber et al., 2023; Kumar et al., 2023; Peralta et al., 2025; Chen et al., 2024; Yan and Geng, 2024). Various studies have examined this topic and yielded differing conclusions. For instance, research conducted in China highlighted changes in heart rate among the elderly between 22 °C and 28 °C (Madaniyazi et al., 2016). Another study noted a significant increase in thermal discomfort at 30 °C when noise levels rose from 55 dB(A) to 85 dB(A); this increase in noise also correlated with a notable rise in heart rate (Guan et al., 2020). Furthermore, a study exploring the impact of noise levels on elderly heart rate found that heart rate increased even at relatively low noise levels of 65 dB(A) (Kraus et al., 2013). Research focusing on traffic noise has shown that higher noise levels (65.6 dB(A)) significantly affect heart rate in the elderly (Huang et al., 2013). The present study supports these findings, indicating that both noise level and ambient temperature markedly influence heart rate in elderly individuals. Consistent with previous research, our study highlights that an increase in noise level, particularly at 30 °C, leads to a rise in heart rate.

The results of this study indicate that as noise levels increase from 55 dB(A) to 85 dB(A), heart rate also rises, particularly at higher ambient temperatures. For instance, at 20 °C, the heart rate increases from 65 beats per minute to 70 beats per minute. At 25 °C, it varies between 65 and 75 beats per minute. At 30 °C, it can reach between 80 and 85 beats per minute. Music tends to cause a lower heart rate compared to the other two types of sounds, even at higher noise levels (remaining below 75 beats per minute at 85 dB(A)). In contrast, traffic and crowd noises lead to a significant increase in heart rate, especially at higher noise levels. For example, loud noises can raise the heart rate to 80 beats per minute at 85 dB(A). Higher ambient temperatures generally result in increased heart rate, with traffic and crowd noises showing more pronounced effects compared to music. At 30 °C, traffic noise can increase heart rate from 85 beats per minute, while crowd noise can increase heart rate between 80 and 85 beats per minute. Thus, this study shows that high noise levels and ambient temperatures generally lead to an increase in heart rate in older adults. Traffic and crowd noises cause a significant increase in heart rate, particularly at 30 °C, whereas music tends to maintain a lower heart rate even at higher noise levels.

The findings of this study are consistent with a study by Meade et al. (2024a, 2024b), which explored the effects of ambient temperature on heart rate in older adults. This study demonstrated that higher ambient temperatures are linked to increased heart rate, particularly when ambient temperatures exceed 26 °C. Specifically, heart rates were found to rise at ambient temperatures of 31 °C and 36 °C, while no significant increase was noted at 26 °C compared to a control ambient temperature of 22 °C. These results indicate that prolonged exposure to elevated indoor ambient temperatures could elevate heart rates in older adults, raising concerns about cardiovascular health. Therefore, it is advisable to keep indoor ambient temperatures below 26 °C to help prevent heat-related complications and reduce mortality risk in this population. Also, the results of our research also align with the findings of the study by Meade et al. (2024a, 2024b), which indicates that exposure to high ambient temperatures (31 °C and 36 °C) leads to an increase in heart rate in older individuals.

The results of the current study are consistent with the findings of Ohlendorf et al. (2023), which explored the effects of musical sounds on the heart rates of elderly individuals. Their research indicated that the type of music had a minimal impact on older adults' heart rates, which tended to be more stable and analogized to those of younger adults. On average, the heart rates of older individuals were consistently lower than those of younger individuals across all sound conditions. Furthermore, the variability in heart rates among older adults was less than that observed in younger adults, suggesting that older individuals may be less sensitive to music and sound stimuli. Altogether, these findings indicate that listening to music does not significantly influence the increase or decrease in heart rates among the elderly. Moreover, in another study conducted by Kraus et al. (2013), examining the effect of noise levels on the heart rate of the elderly, the results showed that even at relatively low noise levels (65 dB(A)), an increase in heart rate was observed, which is not consistent with the findings of our study. This discrepancy may be due to differences in the locations of the studies reviewed and also the duration for which the participants examined.

Previous studies have identified heart rate variations at different ambient temperatures and moderate noise levels (65 dB(A)), with traffic noise having a particularly strong effect (Huang et al., 2013; Guan et al., 2020; Kraus et al., 2013; El Aarbaoui and Chaix, 2020; Cai et al., 2021). Specifically, our study shows that traffic and crowd noise are responsible for the greatest increase in heart rate, reaching 80 to 85 beats per minute at 85 dB(A), underscoring the significant role of environmental factors in elderly care. The present study demonstrates that both ambient temperature and noise level significantly affect heart rate in elderly individuals. Specifically, ambient temperature has a notable impact on heart rate, especially at 30 °C. Additionally, an increase in noise level from 55 dB(A) to 85 dB(A) corresponds with a rise in heart rate, particularly at 30 °C. Traffic and crowd noises lead to the most

[2] The necessary explanations, along with the relevant tables and charts, are provided in Appendix A.

substantial increase in heart rate, reaching between 80 and 85 beats per minute at 85 dB(A).

5. Conclusion

This study examined the impact of ambient temperature, sound type, and noise level on the heart rate of 216 healthy elderly men. The experiments were conducted at three ambient temperature levels (20, 25, and 30 °C) and included four noise levels (55, 65, 75, and 85 dB(A)), as well as three types of sound (music, traffic, and crowd noise). These findings suggest that not only the loudness but also the nature and unpleasantness of sound can be crucial in determining physiological responses. The results of this study hold significant implications for the health and well-being of the elderly, particularly for those living in urban environments with high levels of noise pollution. These findings indicate that prolonged exposure to high noise levels and elevated ambient temperatures increases heart rate, emphasizing the importance of implementing noise control measures and heat management strategies to mitigate these risks. Overall, this research enhances our understanding of the complex relationships between the type of sound and noise level, ambient temperature, and heart rate in elderly individuals, highlighting the need to consider these factors in public health policies and urban planning.

CRediT authorship contribution statement

Mohammad Tohidi: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Karen Fatahi:** Writing – review & editing, Writing – original draft, Validation, Resources, Investigation, Data curation, Conceptualization. **Maryam Beigi:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Funding acquisition.

Consent for publication

Not applicable.

Ethics approval and consent to participate

This study was approved by the Ethics Committee of the Islamic Azad University, Ilam Branch, under the ethics [code No. IR.IAU.ILAM.RES.1403.012].

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.exger.2025.112829.

Data availability

Data will be made available on request.

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